



Positioning in a collaboration network and performance in competitions: a case study of Kaggle

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Abstract

Online innovation competitions are ecosystems where institutions source numerous solutions from knowledge workers through a platform intermediary. By considering how an individual competitor's performance varies based on their social positioning in a competition ecosystem's collaboration network, we illustrate the value of social networks for individual outcomes in online competitions. The study reports results from Kaggle, a popular online competition platform for data science, where a sample of 350,956 users participated in 2,789 competitions over 4 years. We investigate how the number of collaborations, membership in the largest connected component in the network, and diversity of collaboration experiences impact the points and medals earned and how quickly competitors earn their first medal. Results show that positioning has a positive relationship with performance in competitive ecosystems. Relevant to the future of work, the study considers how knowledge workers in future workplaces should manage their online collaborations.

Lay Summary

By competing in online innovation competitions, knowledge workers can gain recognition for developing quality solutions to problems provided by institutions. In recent years, team collaboration has increasingly become more prevalent in such competitions. As a result, studying the impact of a competitor's positioning in the collaboration network helps explain performance differences among competitors, where some competitors may collaborate more than others. In the current study, we investigate the number of collaborations, membership in the largest connected component (LCC) in the network, and diversity of collaboration experiences. With the three variables of interest, we determine the relationship between the collaboration network and the performance of individual competitors. As we continue to prepare for the future of work, it is important to explore the ways that incentivizing collaboration impacts individual experiences when workers are less frequently embedded within the same organizations and are working remotely. These workers will be motivated by their needs and goals, but they will also be susceptible to the relationships they establish during work and will benefit from collaboration.

Keywords: collaboration networks, virtual teams, online competition, data science, online communities

Institutions of various types now commonly engage with innovation competitions to benefit from external expertise, knowledge sharing, and problem-solving (Boudreau & Lakhani, 2013; Majchrzak & Malhotra, 2020). Communication among competitors is common, and ecosystems have become more collaborative and transparent to produce better solutions (Boudreau & Lakhani, 2013; Jin et al., 2021; Machado et al., 2021; Tausczik & Wang, 2017). In competition ecosystems, communication is often asynchronous and mediated through user-generated content. Competitors can manipulate and extend others' artifacts and documents (e.g., software code, reports, and visualizations) while attempting to earn rewards (e.g., monetary compensation, job interviews, and badges). For example, for competitors in Kaggle, the platform also facilitates knowledge sharing and community engagement (Cheng & Zachry, 2020; Tausczik & Wang, 2017). As a result, the competition ecosystem serves as both a hub for collecting solutions and a social media platform where participants engage with each other's work and collaborate.

Collaboration in a competition ecosystem represents how members of distributed labor pools socially interact within platforms. Exploring how individuals are incubated,

developed, cultivated, and arguably exploited helps illuminate how individuals engage with institutions through new platforms and organizing mechanisms in the future of work (Gray & Suri, 2019; Lampinen et al., 2018). Knowledge work is currently shifting toward competitive performances in online communities to gain recognition and signal competence in the labor market and enhance employment opportunities. For example, active participation and recognized contribution to the question and answer community, Stack Overflow, are both linked to job seeking behaviors and expertise signaling of community members seeking career advancement—demonstrating the connection between online community engagement and the labor market (Lee et al., 2022; Xu et al., 2020). Relevant to the current study, Kaggle competitions serve as spaces to develop skills, reputation, and relationships while providing resources for data science work, such as cloud computing, large-scale datasets, and coding examples from competitors.

Competition in online spaces is integral to the future of work. For example, social media influencers and the creator economy demonstrate how attention and engagement from audiences can create economic opportunities for content producers beyond traditional media institutions (Cunningham & Craig, 2021;

Jenkins et al., 2013). Platforms like Kaggle function as professional social media for the future of work, where user engagement and contribution both generate value for the hosting platform and external institutions seeking to solve problems and recruit talent based on performances. Such value matches a key criterion for social media, in that “the value (i.e., benefit or enjoyment) of using social media is derived from the contributions from or interactions with other users rather than content generated by organization or individual hosting the medium” (Carr & Hayes, 2015, p. 51). If knowledge workers of the future will be working on platforms, then online collaborations, competitions, and work practices will likely be situated in social media and online communities.

Kaggle and similar platforms encourage participation by providing ways to seek status within the community (Lampel & Bhalla, 2007; Levina & Arriaga, 2014). Competitors earn distinction through visible signals of competence and skills (e.g., badges, points, and rankings). Additionally, sharing knowledge, collaborating, and engaging with others contribute to gaining status and recognition within the community. Some contributors gain feedback and recognition from peers, which encourages continued participation in online communities (Burtch et al., 2022; Hsieh et al., 2022; Huang et al., 2018; Yan et al., 2023).

Online communities are sources of knowledge production, innovation, and problem-solving that rely on communication among members (Kraut & Resnick, 2012). Collaboration is a common component of competition ecosystems with teams serving as the organizing units responsible for generating novel solutions (Dissanayake et al., 2019; Huang et al., 2022; Quaranta et al., 2022). When competitions encourage collaboration, a network of team memberships emerges over time as community members acquire experience addressing numerous problems. The structure of relationships in collaboration networks may have implications for the productivity and viability of the broader community.

Social networks often exhibit preferential attachment, where some entities have more connections (higher degree) and therefore more access to resources, information, engagement, and attention than others (Johnson et al., 2014; Stephens et al., 2016). This “rich get richer” phenomenon also applies to collaborative settings, where some potential teammates attract more attention and have more options for collaboration (Gómez-Zarà et al., 2019; Twyman et al., 2022; Twyman & Contrará, 2019). Individuals with multiple collaborators can reap benefits, such as access to diverse information, exposure to various working styles, and the ability to innovate (O’Leary et al., 2011; Uzzi & Spiro, 2005; Uzzi et al., 2013).

The current study examines how a competitor’s positioning in a large collaboration network relates to their performance and recognition in the Kaggle competition ecosystem. We analyzed data from 350,956 users who participated in 2,789 competitions, either independently or in teams from 2016 and 2020. Results show that prolific collaborators have more opportunities to produce solutions that attract recognition, highlighting the importance of relationship building in remote collaborations for future workers.

Online collaboration networks

Online collaboration is crucial for many areas of society and promises to continue to grow in importance for the future of

work. In recent years, online teams have been essential for maintaining organizational productivity while working remotely (Yang et al., 2022), developing software (Hahn et al., 2008; Murić et al., 2019), and creating solutions to analytics problems (Dissanayake et al., 2019; Tausczik & Wang, 2017). Prior research has shown that virtual teams must establish relationships through curated profiles, asynchronous coordination, and intentional communication management (Cummings & Dennis, 2018; Gilson et al., 2015; Walther & Bunz, 2005). Because online collaborations leave digital traces of communication, social network analysis helps assess the relevance of competitors’ positioning among one another in platforms. As one joins more teams and collaborations, they gain exposure to diverse working styles and information while contributing knowledge for different outcomes (Akşin et al., 2021; Cross & Cummings, 2004; O’Leary et al., 2011). Collaborating in more teams increases performance outcomes, but not all competitors receive equal opportunities for collaboration in online settings.

Some skills and abilities are more scarce than others, which makes some potential collaborators more desirable and in-demand than others (Gómez-Zarà et al., 2019). Additionally, those who have familiarity with others through pre-existing (and commonly offline) relationships receive more invitations to join teams (Twyman et al., 2022). In remote online settings, other mechanisms beyond the collaboration network may influence individual performance, but prior literature suggests having more collaboration ties is positively associated with performance. Network measures, which reflect relational interactions and social structures in work settings, often serve as a stronger predictor of performance than other attributes (Ahuja et al., 2003; Cross & Cummings, 2004; Manata, 2019). Connecting with others is essential for knowledge workers to access the information needed to perform effectively, and a lack of connections can pose challenges for finding new knowledge online (Robert et al., 2008; Yang et al., 2022). Our first hypothesis serves as a baseline to establish the usefulness of collaborating.

Hypothesis 1: There is a positive association between the number of collaboration ties and performance; performance will improve as the number of collaboration ties increases.

Being centrally embedded in a network impacts an individual’s behavior as they attempt to maintain a favorable reputation among others (Coleman, 1988; Granovetter, 1985; Gulati, 1995; Uzzi, 1997). Embeddedness is defined as the number of common connections held by two actors in a network (Easley & Kleinberg, 2010, p. 55). With this definition, embeddedness extends the conceptualization of network positioning by providing additional information regarding how third parties and redundant relationships correlate with access to resources in a network (Burt, 2005; Halevy et al., 2019; Reagans et al., 2015). As a result, structural considerations in networks, including embeddedness, have been linked to various outcomes, including increased recognition, innovation, creativity, and competitiveness in markets (Chen et al., 2022; Petryk et al., 2022).

Online competitors embedded within larger networks can gain social capital and establish trust based on their engagements and performance (Aral & Walker, 2014; Ellison et al., 2007, 2014). In online communities, socially central members

tend to be more active, attract more attention to their contributions, and assume leadership roles (Halgin et al., 2020; Johnson et al., 2015; Preece & Shneiderman, 2009; Safadi et al., 2021; Tofalvy & Koltai, 2021). For those on the periphery of a network, accessing information, sharing knowledge, gaining social support, and identifying opportunities can be difficult, regardless of their expertise (Safadi et al., 2021; Singh, 2005; Singh et al., 2010). However, they also provide an audience for consumption and help a community interpret and frame its content (Bennett et al., 2018; Tausczik et al., 2018).

When a user is embedded near the center of an online network, they are close to other productive users who contribute greatly to the broader network. Investing energy into building relationships and establishing a reputation with well-connected individuals helps one become more likely to receive access to information and opportunities that others would not learn about otherwise. As a result, online communities can exhibit patterns of behavior similar to ingroups and outgroups from offline settings, such as organizations where employees in subsidiaries may perceive exclusion compared with those at the headquarters (Cristea & Leonardi, 2019). We expect to observe comparable distinctions among competitors in an online collaboration network.

Hypothesis 2: Being embedded within the central hub of a collaboration network is positively associated with performance.

Diversity of collaboration experiences

Collaborating with multiple partners and being centrally positioned in the collaboration network will both be positively related to performance, but the allocation of engagement among collaborators is another factor to consider. Depending on the allocation, a competitor will either collaborate with the same partners repeatedly or choose to work with new partners, which will impact their experiences during collaboration. Establishing familiarity with teammates is a benefit of repeat collaborations (Twyman et al., 2022). This is analogous to forming strong ties within a collaboration network, where individuals build a history of shared experiences. Shared history in a relationship is beneficial because team members are familiar with coordination norms and have common grounding which can easily be updated during the initial stages of renewing the relationship (Espinosa et al., 2007; Gulati, 1995; Levin et al., 2011).

On the other hand, choosing to work with new collaborators allows for weaker ties in the collaboration network to be utilized, which can be helpful for information sharing, professional development, and introducing new perspectives (Aral, 2016; Granovetter, 1973). Collaborating with weaker ties may sacrifice some benefits of familiarity among team members, but there are opportunities for better outcomes. For example, reconfiguring the collaboration network and creating new team combinations increase the quality of work products and members' engagement in virtual organizations (Salehi & Bernstein, 2018).

However, teams must integrate the diversity of perspectives and knowledge to realize their performance potential (Barley et al., 2017; Zaggli & Pottbäcker, 2021). Essentially, weaker ties need to be integrated into the preexisting relationships. Therefore, if a competitor equally allocates their

collaborations among both their strong and weak ties, then they can take advantage of the diversity within their collaboration network. Actualizing benefits from team diversity requires spending time working with less frequent collaborators. By reducing the difference between strong and weak ties, a competitor should benefit from a larger pool of collaborators and a more diverse set of experiences.

Hypothesis 3: There is a positive association between performance and allocating collaborations equally across partners.

Data and methods

Research context

The study focuses on Kaggle (<https://www.kaggle.com>), a platform for data science competitions and learning. At time of data collection, it had over 5 million registered users who can participate in competitions individually or in teams to develop solutions for machine learning and analytic problems. Kaggle has a progression system that measures a user's productivity and contribution to the community based on four activities: dataset sharing, discussion, notebook (script) sharing, and competition. Users earn medals and points based on their feedback and rankings from the different activities. A detailed description of the four types of progression can be found on the Kaggle website: <https://www.kaggle.com/progression>.

There are two types of Kaggle competitions: "in-class" and "ranked." "In-class" competitions are for skill development and practice but do not offer rewards. On the other hand, participants in "ranked" competitions have their performances ranked based on the accuracy of their models and are rewarded with points and medals. In both types of competitions, participants can work independently or form teams with other competitors.

Data sample

The current study uses the "Meta Kaggle" dataset collected from Kaggle (<https://www.kaggle.com/kaggle/meta-kaggle>), updated daily and documenting user activities from Kaggle's inception until September 17, 2020. The dataset includes user information, team collaborations and performances, and competition details. The user data include the account creation date, competition participation, and progression—including their points and medals from dataset, notebook, discussion, and competitions—in the platform. Competitions data contain the start dates, deadlines, team memberships, and team performances.

Kaggle introduced the medal rewarding system on July 15, 2016, as indicated by the platform metadata provided by the dataset. Competitions with a submission deadline before that date distributed medals on an *ad hoc* basis over time ($N=335$, with 156 ranked competitions). After July 15, 2016, there were 2,789 competitions, of which 119 of them were ranked. The study data contained 2,350,759 teams formed by 929,560 distinct users participating in 2,789 competitions, of which 116,974 teams were formed by at least two users. 621,672 competitors participated in at least one ranked competition. Table 1 presents some notable statistics about the dataset.

Table 1. Competitions, competitors, and teams by year after July 15, 2016

Year	2016	2017	2018	2019	2020	Total
Number of competitions	63	278	585	964	899	2,789
Number of ranked competitions	9	25	26	37	22	119
Number of competitors	108,279	353,002	378,752	347,128	272,299	929,560
Number of teams (team size > 0)	124,933	527,684	644,808	680,623	372,531	2,350,579
Number of teams (team size > 1)	4,801	15,269	25,454	42,870	28,580	116,974

Collaboration network analysis

We describe the collaboration network from which our treatment variables are derived. To extract the relevant network variables, we constructed a weighted undirected network from all competitors in both “in-class” and “ranked” competitions since Kaggle’s inception. Nodes are the competitors, and they are linked by an edge when competitors are members of the same team in at least one competition. The weight of the edge is the number of collaborations between two competitors.

We obtained a sparse network from the collaborations during all competitions with more than 1 million nodes (including competitors who did not finish competitions) but less than 135 thousand edges (density = 1.696×10^{-7}). From this network, we extracted more than 1 million connected components. In social networks theory, a component of an undirected graph is a connected subgraph that is independent from any larger connected subgraph. The LCC is the connected subgraph with the highest number of nodes (Newman, 2018). There are 9,316 nodes (0.7% of all competitors) and 26,668 edges in the LCC, which has a higher density (6.15×10^{-4}) than the complete network. Table 2 contains summary statistics of the collaboration network and LCC.

Statistical analysis

To test the hypotheses, we conducted a series of *t*-tests on matched samples of users following the Propensity Score Matching (PSM) method. We split competitors into *Control group* and *Treatment group* based on three treatment variables: *Number of collaborators*, *Belonging to LCC*, and *Collaboration Diversity*, using the appropriate cut-off values. We then measure three performance metrics as dependent variables: *Points*, *Medal*, and *Medal tenure*.

Propensity score matching

We used PSM, an analytic technique that evaluates the effect of treatments on groups with balanced confounding variables (Caliendo & Kopeinig, 2008). PSM uses propensity scores of each observation to match any observation in the treatment group to one or more observations in the control group. The propensity scores are computed by a logit model with confounding variables. For the analysis described below, we performed one-to-one matching of observations in the *Treatment* and *Control* groups by matching competitors with the closest propensity scores. For hypothesis testing, we ran *t*-tests to compare *Treatment* and the new *Control* groups on the dependent variables.

Dependent variables

User performance in competitions is measured by three variables: (a) *Points*, indicating the total points received in all the competitions; (b) *Medals*, representing the total number of

Table 2. Summary statistics for the team collaboration network

	Complete Network	LCC
Number of nodes	1,261,673	9,316
Number of edges	134,966	26,668
Density	1.696×10^{-7}	6.15×10^{-4}
Avg. clustering coefficient (weighted)	1.59×10^{-3}	0.024
Transitivity	0.661	0.493
Number of connected components	1,187,665	NA
Diameter	Unconnected	27
Radius	Unconnected	14
Avg. shortest path (weighted)	Unconnected	8.11

medals achieved in competitions; and (c) *Medal tenure*, defined as the time in days between a user registering on Kaggle and achieving their first competition medal. Since Kaggle introduced medals on July 15, 2016, calculating *Medal tenure* for users who had participated in any ranked competitions beforehand is unreliable. Also, the metadata from Kaggle did not provide information about *Points* in competition for users who registered after October 18, 2019. To account for these limitations, we only analyzed a subset of users who registered on Kaggle before October 18, 2019 and competed in their first ranked competition with a submission deadline after July 15, 2016 ($N = 350,956$). Table 3 shows the descriptive statistics of all key variables and Table 4 shows correlations.

Treatment variables

We designed three binary treatment variables based on social network measures of competitors and their collaborations from the aforementioned collaboration network in Kaggle. The three variables are the *Number of collaborators* (*collaborators*), *LCC membership*, and *Collaboration diversity* (*diversity*).

Number of collaborators (*collaborators*) captures information about the quantity of connections and is the total number of users a competitor has collaborated with in competitions. We converted the number of collaborators into a binary variable to distinguish between competitors who only work independently (*Collaborators* = 0) and those who collaborate with at least two other competitors. Based on the degree distribution of the complete collaboration network, the cutoffs correspond to the 5th and 95th percentiles. The cutoff is also meaningful based on the calculation of *Collaboration Diversity* (described below), which requires a minimum of two collaborators. Competitors with *Number of collaborators* ≥ 2 are the *Treatment group*, while competitors with zero collaborators are the *Control group*.

LCC membership differentiates between competitors who are positioned within the LCC of the collaboration network and the competitors who are not. LCC captures information about the quantity of direct and indirect connections. Competitors in the LCC are the *Treatment group* and are

Table 3. Descriptive statistics of key variables on competitors ($N = 350,956$)

	Mean	SD	Min	25%	Median	75%	Max	Missing
Dependent variables								
Points	68.28	741.02	0	0	0	0	120,933	0
Medal	0.038	0.340	0	0	0	0	33	0
Medal tenure (days)	403.47	423.58	2	91	266	575	3,269	342,250 ^a
Independent variables								
Number of collaborators	0.28	1.08	0	0	0	0	50	0
Number of collaborators (for competitors with at least one collaborator)	2.39	2.20	1	1	2	3	50	309,284 ^b
LCC membership	0.009	0.096	0	0	0	0	1	0
Collaboration diversity (for competitors with at least two collaborators)	0.687	0.021	0.289	0.693	0.693	0.693	0.693	327,011 ^c
Controls								
Number of competitions	2.77	3.51	1	1	2	3	158	0
Ranked proportion (of competitions)	0.850	0.228	0.031	0.667	1	1	1	0
Tenure (days)	987.72	494.90	337	619	898	1,248	3,865	0

^a Only 8,706 competitors have won at least one medal.^b Only 41,672 competitors have at least one collaborator.^c 23,945 competitors have at least two distinct collaborators.**Table 4.** Pairwise Spearman's correlations between key variables

	1	2	3	4	5	6	7	8
1. Points	1							
2. Medal	0.40	1						
3. Medal tenure	0.26	0.05	1					
4. Number of collaborators	0.39	0.27	-0.05	1				
5. LCC membership	0.19	0.31	0.02	0.28	1			
6. Collaboration diversity	-0.21	-0.17	-0.11	0.07	-0.09	1		
7. Number of competitions	0.30	0.15	0.35	0.14	0.09	-0.31	1	
8. Ranked proportion (of competitions)	-0.06	-0.02	-0.25	-0.05	-0.02	0.11	-0.65	1
9. Tenure	0.08	0.03	0.50	0.03	0.01	0.00	0.11	-0.05

All correlations are significant at $p < .001$.

more embedded since their collaborators have likely collaborated with others or have indirect relationships through other competitors. Meanwhile, competitors not positioned in the LCC are the *Control* group.

Collaboration diversity quantifies the diversity of collaboration experiences for each competitor. To calculate the *Collaboration diversity*, we use the set of a competitor's collaborators and calculate the normalized Shannon's entropy index for the set, which is a common measure of diversity in terms of a group's functional background (Harrison & Klein, 2007), and the randomness in motion and diffusion rate in various fields (Cincotta et al., 2021).

To compute the Shannon's entropy index for a single competitor, we extracted the list of the competitor's collaborators in all competitions, with repetitions. Allowing repetitions, the same collaborator can appear in the list multiple times. The more frequently a competitor collaborated with the user, the more frequently the collaborator will appear in the list. For a user u who has the set of collaborators V , the Shannon's entropy was computed as $H(u) = \sum_{v \in V} P(v) \log_2 P(v)$, where V stands for the set of all distinct collaborators of u and $P(v)$ indicates the probability u and v collaborated. Shannon's entropy, similar to Blau's index or Gini coefficient, is useful for quantifying the diversity of collaboration experiences.

To account for the maximum value of Shannon's entropy index, which is predetermined by the size (N) of collaborator set V (Harrison & Klein, 2007), we normalized the variable by its maximum $\log_2 N$. The normalization (between 0 and 1)

ensures that the value of this measure is comparable between competitors with varying numbers of collaborators. Regardless of the number of distinct collaborators, a higher value indicates a more balanced, diverse, and less predictable set of collaborators for a competitor. We only computed this variable for competitors with at least two distinct collaborators ($N = 309,284$).

Using this formulation, we separate the competitors that are inclined to repeat collaborations with only a few of their collaborators (lower entropy, not diverse; reliant on strong ties) from those with less predictable collaboration patterns (higher entropy, diverse; minimal distinction between strong and weak ties). We split competitors based on 5th percentile and 95th percentiles, which still retained 92% of the users with at least two distinct collaborators. Competitors with *Collaboration diversity* equal to or higher than the upper 95th percentile are the *Treatment* group, while users equal to or lower than the lower 5th percentile are the *Control* group. Table 5 details the *Treatment* and *Control* groups with the number of observations in the matched samples.

Control variables

The control variables in PSM control for differences between the groups of users. We use the following control variables: (a) *Competition count*, the number of competitions a user had participated in before September 17, 2020; (b) *Ranked proportion (of competitions)*, the proportion of competitions which provides rewards for winners and counts toward the

Table 5. Matched samples in research design

Treatment	DV	Cutoff _C	Cutoff _T	N _C	N _T
Number of collaborators (Collaborators)	Points	Collaborators = 0	Collaborators ≥ 2	23,945	23,945
	Medals	Collaborators = 0	Collaborators ≥ 2	23,945	23,945
LCC membership (LCC)	Medal Tenure	Collaborators = 0	Collaborators ≥ 2	3,039	3,039
	Points	LCC = 0	LCC = 1	3,235	3,235
	Medals	LCC = 0	LCC = 1	3,235	3,235
	Medal Tenure	LCC = 0	LCC = 1	1,687	1,687
Collaboration diversity (Diversity)	Points	Diversity ≤ 0.65	Diversity ≥ 0.69	1,247	1,247
	Medals	Diversity ≤ 0.65	Diversity ≥ 0.69	1,247	1,247
	Medal Tenure	Diversity ≤ 0.65	Diversity ≥ 0.69	505	505

total points of users; (c) *Tenure*, the difference in days between user registration on Kaggle and date of data collection (September 17, 2020); and (d) *Number of collaborators*, the number of distinct collaborators a user has worked with in a team during the tenure.

Results

Descriptive statistics

The sample includes all competitors who registered before October 18, 2019, and participated in their first ranked competition with a submission deadline after July 15, 2016, when the medals were introduced. Based on the collected dataset, this sample of competitors had consistent medal award dates and points from competitions. Table 3 shows the descriptive statistics of the key variables at the competitor level ($N = 350,956$).

Since many competitors in our sample have zero points ($N = 289,623$; 82.5%) and have never won any medals ($N = 342,250$; 97.5%), the distributions of these two variables are extremely skewed—only a few competitors obtained high points and won many medals. For those who have won at least one medal ($N = 8,706$; 2.5%), it took them on average more than a year (Mean = 403.47; $SD = 423.58$) to win the first medal after they registered on Kaggle. Most competitors have not worked with any collaborators ($N = 309,284$; 88.1%). Additionally, collaborating with multiple other competitors is relatively rare; 23,945 competitors (6.8%) in the sample have worked with at least two distinct collaborators.

The distribution of *Competition count* is also skewed with 46% of competitors only participating in one competition ($N = 161,352$). Among all the competitors, 229,269 competitors only participated in ranked competitions (65.3%). On average, the sample competitors have a tenure within Kaggle of more than two and a half years (Mean = 987.72, $SD = 494.90$).

We provide pairwise Spearman's correlations between key variables in Table 4. The correlation values are computed with complete observations (without missing values) on the variables. All pairs of correlations are significant at $p < .001$. Notable positive correlations are present among the dependent variables, except between *Medal tenure* and *Points*. The *Number of collaborators* and *LCC membership* were positively related to *Points* and *Medals*. Weaker correlations between *Medal tenure* and the *Number of collaborators* (-0.05) and *LCC membership* (0.02) exist. *Collaboration diversity* was negatively correlated with all three dependent variables. Note that a negative correlation with *Medal tenure* indicates a

shorter time for a competitor to win their first medal, which indicates positive performance.

Hypothesis testing

Using PSM, we evaluated the associations between the three treatment variables and three measures of competitor performance. The three treatment variables were *Number of collaborators*, *LCC membership*, and *Collaboration diversity*. The number of *Points*, number of *Medals*, and *Medal tenure* represent competitor performance. For each treatment variable, we matched each competitor in the *Treatment* group with the competitor from the rest of the sample who possessed the closest propensity score to create a *Control* group with the same distribution of confounding variables, including *Competition count*, *Ranked proportion (of competitions)*, *Tenure*, and *Number of collaborators* (for *LCC membership* and *Collaboration diversity*). Table 6 presents the main results that test our hypotheses and are explained below.

Hypothesis 1: The number of collaborators and performance

When testing Hypothesis 1, we found that collaborative competitors (*Collaborators* ≥ 2) perform better than solo competitors according to all three dependent variables. Collaborative competitors have more *Points* ($t = -20.61$, $p < .001$) and *Medals* ($t = -36.74$, $p < .001$) than solo competitors. The results are significant and the differences in means are substantial; collaborative competitors achieve an average of 430.28 points, which is more than four times greater than the average number of points obtained by solo competitors (100.48). Similarly, the collaborative competitors earned seven times as many medals as solo competitors (0.304–0.043 medals, on average). Only 1.4% of solo competitors have a medal, compared with 29.3% of collaborative competitors. Note that the size of the *Treatment* and *Control* groups was equally balanced with 23,945 competitors in each group.

When comparing the competitors who won at least one medal, collaborative competitors earn their first medal, on average, 166 days faster than solo competitors ($t = 17.52$, $p < .001$). It takes 428.82 days for a solo competitor to win their first medal, while collaborative competitors earn their first medal in 252.54 days, on average. For this analysis, we compared 3,039 competitors in each group. Overall, there is support for Hypothesis 1.

Hypothesis 2: LCC membership and performance

There is partial support for Hypothesis 2 based on analysis of 3,235 competitors in each group. The competitors who are LCC members in the collaboration network ($LCC = 1$),

Table 6. Comparisons of competitor performance in *Treatment* and *Control* groups

Treatment	DV	Mean _C	SD _C	Mean _T	SD _T	Cohen's <i>d</i>	<i>T</i> statistic	Confidence interval	<i>df</i>	<i>p</i>
Number of collaborators	Points	100.48	778.34	430.28	2,350.17	0.188	-20.61	[-361.16 to -298.44]	29,134	<.001
	Medal	0.043	0.356	0.304	1.042	0.336	-36.74	[-0.275, -0.248]	29,445	<.001
	Medal tenure	428.82	445.86	262.54	273.66	0.450	17.52	[147.68-184.89]	5,042.5	<.001
LCC membership	Points	512.20	1,787.32	2,009.70	5,886.75	0.344	-13.85	[-1,709.56, -1,285.43]	3,825.2	<.001
	Medal	0.356	0.961	1.226	2.34	0.486	-19.55	[-0.957, -0.782]	4,293.5	<.001
	Medal tenure	337.45	343.64	404.44	397.30	0.275	-5.24	[-92.06, -41.91]	3,303.4	<.001
Collaboration diversity	Points	1,900.57	5,824.20	50.68	174.44	0.449	11.21	[1,526.17-2,173.61]	1,248.2	<.001
	Medal	1.192	2.55	0.144	0.354	0.575	14.37	[0.905-1.191]	1,293.9	<.001
	Medal tenure	450.84	403.93	296.96	428.84	0.369	5.87	[102.44-205.32]	1,004.4	<.001

garner more *Points* ($t = -13.85$, $p < .001$) and earn more *Medals* ($t = -19.55$, $p < .001$) on average than those who are not in the LCC. The non-LCC competitors accumulate 512.20 points while the LCC competitors obtain 2,009.70 (close to four times as many) points, on average.

The comparison of *Medal tenure* between competitors within and outside the LCC showed that non-LCC competitors earned medals faster ($t = -5.24$, $p < .001$). Non-LCC competitors earned medals approximately nine-and-a-half weeks faster (67 days) than LCC competitors. This comparison was based on a sample of 1,687 competitors in each group. Overall, *LCC membership* is positively associated with higher performance rankings in competitions, but competitors are slower in earning their first medal.

Hypothesis 3: Diverse collaboration experiences and performance

Hypothesis 3 suggests that allocating collaborations equally among a set of collaborators is better for performance than concentrating collaborations among a subgroup of the set. However, the results are mixed. Collaboration diversity is negatively associated to the number of *Points* ($t = 11.21$, $p < .001$) and *Medals* ($t = 14.37$, $p < .001$), but it is positively associated with a shorter *Medal tenure* ($t = 5.87$, $p < .001$). The high-entropy competitors who maintain a relatively diverse collaboration pattern have a lower average number of *Points* (50.68) and *Medals* (0.14) compared with low-entropy competitors who concentrate on a subset of their collaborators; an average of 1,900.57 points and 1.19 medals, respectively. However, having diverse collaboration experiences is associated with a reduced *Medal tenure*; an average of 296.96 days compared with 450.84 days. The analysis compares groups of 1,247 competitors for *Points* and *Medals*, and groups of 505 competitors for *Medal tenure*. These results partially support Hypothesis 3 and suggest that allocating equally among collaborators is helpful for winning earlier in a competitor's tenure, but concentrating on a subset of collaborators is better for other performance measures.

Discussion

As innovation contests have become ubiquitous over the past decade, these competition ecosystems have created a communication interface for institutions to engage with external knowledge workers through problem-solving activities. A valuable, yet underexplored, component of these interfaces are the relationships that emerge among competitors who address the problems sourced from institutions. The current study illustrates that collaboration networks are helpful

considerations when assessing the performance of individual competitors within the Kaggle competition ecosystem.

We investigated individual positioning within a collaboration network that emerged over 4 years of competitions in Kaggle. Results showed that collaborating with multiple partners (high degree in collaboration network) was positively associated with competitors earning medals and points as well as decreasing the time needed for their first accomplishment. Collaborating in Kaggle competitions is helpful because team members are equally credited and share rewards.

To extend our understanding of the embeddedness of individual competitors within the collaboration network, we assessed the extent to which membership within the network's LCC would be positively associated with performance. After controlling for relevant competitor characteristics, results showed that comparable competitors will experience different performance outcomes based on whether they belong to the LCC. LCC competitors accumulate more points and win more medals than non-LCC competitors, but non-LCC competitors win their first medal earlier in their tenure. Taken together, being embedded within a well-connected community benefits performance, but it is possible to experience positive performance more quickly when not embedded.

The last set of results demonstrates that distributing collaboration experiences among a set of collaborators serves different performance goals. Creating relatively stronger and weaker ties in a collaboration network is positively related to a competitor's performance with respect to their accumulation of awards (points and medals). However, having diversity among collaboration experiences and not repeatedly collaborating with the same subset of collaborators resulted in a shorter time to receive a medal for competitors. The partial support for the hypothesis suggests a tradeoff between collaboration and early achievement. By not concentrating on a subset of collaborators, some competitors were able to achieve early success within the competition ecosystem. From the finding, there may be justification for encouraging newcomers to collaborate with multiple partners and with less repetition in competitive organizations and other ecosystems. However, competitors should concentrate their collaborations around a smaller set of partners to position themselves for positive performance indicators, which would suggest that familiarity is helpful when positive performance is not urgently needed.

Implications for communication research

The utilization of a network perspective for the study of individual performance in an online competition ecosystem contributes to the computer-mediated communication (CMC)

research community. Because the ecosystem contains both individual competitors as well as teams participating in knowledge-intensive problem-solving competitions, networks are a helpful way to account for the interactions that exist within the platform given the interdependence among competitors. Much of the existing scholarship on Kaggle and other competitions platforms highlights either individual or team performance outcomes (Cheng & Zachry, 2020; Dissanayake et al., 2019; Huang et al., 2022; Jin et al., 2021; Sun et al., 2021; Tausczik & Wang, 2017) while our study attempts to reconcile the relationships among teams and individual competitors by focusing on the collaboration network within the ecosystem.

The collaborations within the ecosystem are particularly relevant to CMC due to the role of digital interfaces and artifacts as collaboration spaces; specifically, the programming scripts and discussion forums where competitors engage, collaborate, and compete with one another. We observe that the competition ecosystem manifests social interactions that reflect common interactions that occur in a broad range of online communities and social media platforms, which have been a long-term focus of CMC scholarship. The interactions are centered around competitors who produce and share professional documents and knowledge, which now incorporate elements of social media, broadly defined. For example, while competitive performances are the main activity, performance in a competition ecosystem is complemented by knowledge sharing (Tausczik & Wang, 2017) and receiving feedback from peers (Burtch et al., 2022; Huang et al., 2018). As a result, competitors can actively seek status and accumulate social capital within the ecosystem, reflecting behaviors common to social media where gaining positive reactions from a broader audience is a common goal (Ellison et al., 2007, 2014; Lampel & Bhalla, 2007; Levina & Arriaga, 2014).

As organizations continue to integrate social media into the future of work, competition ecosystems can provide value in at least two different ways; by providing a repository of generated solutions and giving visibility to producers. Specifically, the persistence of content produced for competition allows others to engage with the embedded knowledge, which provides value to an audience that may not have participated in a competition. As a result, competitors receive acknowledgment and capital for their contributions while also helping create spaces for sustained engagement and knowledge sharing in the platform (Majchrzak et al., 2013). For distributed and remote organizations, social media helps develop a shared understanding among members by providing visible expertise indicators through the capture of knowledge contributions and showcasing relationships and interactions among members (Ellison et al., 2015; Leonardi, 2015, 2018).

As knowledge workers equip themselves for a future where they are remote and geographically isolated from their organizations and peers, the current study illustrates the impact of relationships. Communication networks are a core tenet of communication research, and this study contributes to communication network scholarship by showing the associations between performance outcomes and relationship building through collaboration in a competitive online ecosystem. The Kaggle collaboration network emerged over a 4-year period with thousands of competitions and contained many subgroups of competitors. Given the network size and online nature of the context, many competitors would not have

encountered one another. Nevertheless, LCC competitors experienced performance benefits even though they were similar in many ways to the non-LCC competitors, pointing to the importance of positioning in a social network. Maintaining long-distance relationships requires extra effort to signal availability, build awareness, and create shared understanding, but the additional investment helps provide a competitive advantage over isolated individuals who have not been able to build connections with others (Cristea & Leonardi, 2019). If one attempts to compete to their fullest potential, then competing to secure productive and diverse collaborations is a helpful advantage to pursue.

Another contribution illuminates how the diversity of collaboration experiences helps clarify the concept of tie strength in collaboration networks. Depending on the outcome, having a network with an equally distributed set of collaborations is advantageous. By concentrating collaborations and focusing on a subset of ties (strong ties), competitors who prioritize familiarity in collaboration experiences show increased performance, but perhaps sacrifice exposure to different perspectives and ways of collaborating (Akşin et al., 2021; Espinosa et al., 2007; Twyman et al., 2022). However, for individuals in competitive settings, it is also opportunistic to distribute time among as many collaborators as possible to help increase early success. By investing in communication across more collaboration ties, a competitive person can reduce the differences in engagement and attention given to partners, which aids how quickly new competitors can earn their first performance recognition. In the future of work, knowledge workers should be proactive in building as many relationships as possible before they focus their energy on developing strong ties. Weak ties promise to remain valuable as an unchanging constant for experiencing positive economic outcomes well into the future (Aral, 2016; Granovetter, 1973). Early success begets future success, and future workers will need weak ties as they navigate the distributed work environments of the future.

Practical implications

For competition ecosystems like innovation contests and computer programming competitions, it is important to acknowledge that these platforms are social environments where people benefit from information access. Collaboration and knowledge sharing around specific skills and topics help individuals as they participate in competitions (Tausczik & Wang, 2017). While the current study focused on the collaboration network in Kaggle competitions, other networks are informative for understanding the value of such ecosystems. For example, networks that link competitors to the institutions for which they compete will provide information regarding competitor topical interests and preferences as well as how concentrated or distributed solutions are from the community of solvers (Stephens et al., 2016). With this type of information, competitors and institutions can assess indicators of fit and interest based on performance, which does not require formal recruitment and talent acquisition processes.

Limitations

The current study provides a helpful perspective for understanding how susceptible competitors are to their positioning in emergent collaboration networks. However, the study only focuses on a single competition ecosystem, which limits the claims that can be made in at least three ways. The initial

concern is the type of problem typically posed to competitors in Kaggle, which is usually centered on analytical tasks, such as utilizing machine learning and data science methods. These types of problems do not reflect the range of problems that innovation competition ecosystems may encounter. The next issue comes from the design of Kaggle, which encourages and rewards collaboration. It is currently unclear how prevalent formalized team collaborations are in other competition platforms. Lastly, the analysis reported in the current study is centered on the collaboration network but does not address team performance and how team dynamics are related to the network positioning of individual members. These limitations provide additional opportunities to further explore innovation competitions and platforms.

Future research directions

There are at least two additional research areas that warrant future investigation: interrogating other collaboration networks and social processes in different competition ecosystems as well as determining the impact of competition incentives on individual participation and performance. The first area would allow for comparisons to be made between ecosystems and increase generalizability of the findings from this study. Additionally, an investigation of competition incentives will be meaningful for providing explanations of competitor behaviors. For example, in Kaggle, some competitors choose to share aspects of their work. This type of behavior could be occurring in response to a specific set of incentives that Kaggle has established. It is therefore possible for various incentives to encourage both competitive and cooperative behaviors. Deepening our understanding of the role of incentives in competition ecosystems provides a more comprehensive view of online competition.

Another set of future investigations can more rigorously examine problem-solving, team assembly and composition, and knowledge diversity within competition ecosystems. While a strength of the current study lies in its utilization of a network perspective to explain individual performance differences, there is an opportunity to focus on team processes that promote different collective problem-solving and coordination strategies based on the types of knowledge required by a competition. Using Kaggle as an example, the competitions have different types of modeling and prediction requirements, which will require different skillsets from competitors. As a result, when considering the knowledge profiles of competitors, it would be helpful to explore whether some types of competitions are avoided or relatively inaccessible to some population of competitors (potentially learners or newcomers). Also, given the rich history of collaborations in the ecosystem, questions related to team assembly and composition can be explored. For example, which team members initiate collaboration and what role do the most competent team members play in the assembly process? How much communication is needed to build teams and does communication in the assembly process affect team coordination? Additional future research exploring how teams integrate and combine diverse knowledge would be helpful. For example, a competitor could be a specialist in analyzing a specific type of data but needs to collaborate with other specialists for different tasks required by a competition.

Conclusion

Competing online is an integral component of the professional experiences of future workers, who will be developing skills and establishing their reputations in online settings. As we continue to assess the platforms that serve as the infrastructure for the future of work, we need to consider how social media and social networking sites bolster the interfaces between knowledge workers and institutions. Knowledge workers provide value through their generated content and engagement with others in platforms, and instead of treating knowledge workers as isolated and autonomous agents, the findings illustrate that an individual's positioning in a collaboration network impacts their outcomes. From the research on a collaboration network, we uncover multiple ways that an individual's relationships are positively related to performance outcomes and the speed with which they experience success. To remain competitive, the knowledge workers of the future will need to proactively manage their relationships and the diversity of their experiences outside of organizations due to the increasing prominence of remote work in online settings. Such insights will allow for institutions, competition platforms, and knowledge workers to account for the value of establishing collaboration networks to support working arrangements centered around remote and decentralized labor.

Data availability

While Kaggle's data are updated daily, no historical archive is available. The authors can provide the data used in this article from September 17, 2020. Data can be obtained upon reasonable request to the corresponding author.

Funding

This research was supported by the National Science Foundation (IIS-2104551).

Conflicts of interest: The authors declare that there are no conflicts of interest.

Acknowledgments

We thank Nicole Ellison, Nancy Baym, the anonymous reviewers, and the rest of the editorial team for constructive and helpful feedback. Special thanks to Janet Fulk and Peter Monge for their insights during the review process.

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